Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “motor” and “generator” to compare your model or result. Sentence starters are optional.

motor: a device converting electrical energy to mechanical output

generator: a device converting mechanical input to electrical output

efficiency: useful output divided by input

01 Warm-up: bring back an earlier idea

A bar magnet is cut into 3 pieces. Does this produce isolated north and south poles?
Sketch the usual result.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$.
Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives 20 J electrically and delivers 12 J mechanically; 8 J goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp.
Label each input and useful output.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

04 Apply the physics

A hand-cranked generator receives 50 W of mechanical power and supplies 12 V at 3 A. Complete the energy-rate account.
Find electrical output power, efficiency and the rate of transfer to other forms.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A motor drives a generator that feeds the motor.
Explain why real losses prevent the loop from running forever without external input.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Move from r to 4 r away from a point charge.
Compare field magnitude and state the source model assumption.
Label the source charge, distances and electric-field directions.

Word help: charge = a property responsible for electric interactions

Sentence starter: Both ... . They differ in ... because ... .



Source charge Field location

Marzano check - circle: 1 2 3

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Continue down each column.

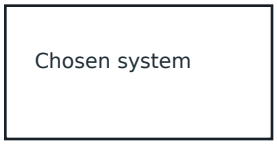
Sentence starters are optional. You may plan in any language.

07 Exit check

A motor is 80% efficient and a generator is 75% efficient. **Find** their combined useful fraction in sequence.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

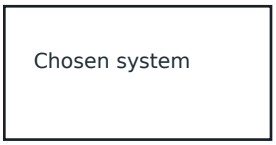
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A motor receives 24 W and operates at 75% efficiency. It lifts a 3 kg load at steady speed with g=10. **Find** the speed and wasted power.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3